

MALEXANDER, Sweden

WMO: 025520

Lat: 58.072N

Lon: 15.233E

Elev: 197

StdP: 98.98

Time Zone: 1.00 (EUC)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-14.8	-12.0	-16.7	0.9	-13.7	-14.2	1.1	-10.6	8.9	5.3	8.0	3.6	1.2	20	0.572

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	9.6	27.3	17.7	25.4	17.1	23.3	16.4	19.3	24.2	18.3	22.9	17.3	21.7	2.7	220

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
17.7	13.0	20.7	16.7	12.2	19.7	15.7	11.4	19.0	55.9	23.9	52.5	22.9	49.5	21.5	22.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 7.0	6.2	5.5	-17.2	29.6	4.0	2.0	-20.1	31.1	-22.4	32.2	-24.6	33.4	-27.5	34.9		
(5)			-17.7	21.0	3.8	1.3	-20.4	21.9	-22.6	22.6	-24.8	23.3	-27.5	24.3		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	6.9	-2.3	-2.2	0.8	5.7	10.5	14.6	17.3	16.1	11.9	6.9	2.8	-0.2	(6)
(7)		DBStd	7.81	4.60	4.55	4.30	3.57	3.37	3.03	3.04	2.68	3.00	3.71	3.74	4.57	(7)
(8)		HDD10.0	1837	383	343	284	135	33	2	0	1	16	108	217	316	(8)
(9)		HDD18.3	4227	641	576	543	378	243	118	58	79	194	355	466	574	(9)
(10)		CDD10.0	698	0	0	0	7	49	140	227	191	72	12	1	0	(10)
(11)		CDD18.3	44	0	0	0	0	1	7	26	11	1	0	0	0	(11)
(12)		CDH23.3	419	0	0	0	0	10	87	247	73	2	0	0	0	(12)
(13)		CDH26.7	79	0	0	0	0	0	13	54	12	0	0	0	0	(13)
(14)	Wind	WSAvg	2.6	2.8	2.8	2.9	2.6	2.6	2.5	2.3	2.3	2.5	2.6	3.0	2.9	(14)
(15)	Precipitation	PrecAvg	588	35	32	29	34	45	67	79	70	53	52	51	41	(15)
(16)		PrecMax	733	68	53	51	78	82	153	141	155	126	128	92	73	(16)
(17)		PrecMin	450	9	4	6	2	7	0	9	18	13	14	15	8	(17)
(18)		PrecStd	73	18	11	11	22	21	36	37	35	33	26	20	18	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	7.1	9.1	14.3	19.8	24.7	28.4	30.5	27.8	23.0	17.0	11.6	10.1	(19)
(20)			MCWB	5.9	6.3	8.0	11.3	15.5	16.9	18.7	18.6	17.0	13.2	10.4	8.8	(20)
(21)		2%	DB	5.6	6.3	11.2	17.2	21.7	26.0	28.3	25.4	20.4	14.5	10.1	8.2	(21)
(22)			MCWB	4.5	4.4	6.4	10.0	13.8	16.7	18.3	17.7	15.5	12.2	9.0	7.0	(22)
(23)		5%	DB	4.5	5.1	8.5	14.7	19.4	23.5	26.3	23.1	18.2	13.1	8.9	6.8	(23)
(24)			MCWB	3.5	3.6	5.1	8.8	12.5	15.6	17.5	17.2	14.3	11.3	8.0	5.7	(24)
(25)		10%	DB	3.3	3.9	6.5	12.4	17.2	21.3	24.2	21.3	16.5	11.9	7.8	5.4	(25)
(26)			MCWB	2.4	2.6	4.1	7.6	11.3	14.4	17.0	16.1	13.5	10.3	6.9	4.4	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	6.3	6.7	8.6	12.7	17.1	19.2	20.8	20.4	18.0	13.8	10.8	9.1	(27)
(28)			MCDWB	7.0	8.1	12.4	19.0	22.5	24.0	26.4	24.6	21.2	15.4	11.3	9.9	(28)
(29)		2%	WB	4.7	4.9	6.9	10.9	15.2	17.5	19.7	19.0	16.1	12.6	9.1	7.1	(29)
(30)			MCDWB	5.4	6.2	10.4	15.3	20.0	23.6	25.1	22.9	19.1	14.2	9.8	8.0	(30)
(31)		5%	WB	3.6	3.7	5.6	9.4	13.6	16.2	18.7	17.9	14.9	11.5	8.1	5.8	(31)
(32)			MCDWB	4.3	4.9	8.0	13.6	17.5	22.2	24.0	21.3	17.6	13.0	8.8	6.6	(32)
(33)		10%	WB	2.5	2.6	4.3	8.1	12.1	15.1	17.7	17.0	13.8	10.5	7.0	4.5	(33)
(34)			MCDWB	3.2	3.7	6.2	11.8	16.0	20.3	22.6	20.3	16.0	11.7	7.8	5.3	(34)
(35)	Mean Daily Temperature Range	5% DB	MDBR	4.2	5.1	7.1	9.7	10.0	9.8	9.6	8.4	7.5	5.7	3.9	3.8	(35)
(36)			MCDBR	4.8	5.6	10.3	13.5	13.6	13.9	13.6	11.3	9.8	6.5	4.7	4.9	(36)
(37)			MCWBR	4.5	4.3	6.5	7.7	7.1	6.7	5.5	5.3	5.5	4.6	4.3	4.6	(37)
(38)		5% WB	MCDWB	4.6	4.9	9.4	11.5	11.4	12.1	11.4	9.6	9.0	6.1	4.5	4.7	(38)
(39)			MCWBR	4.5	4.1	6.3	7.1	6.9	6.4	5.6	5.4	5.5	4.6	4.2	4.6	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.277	0.312	0.331	0.369	0.366	0.355	0.379	0.372	0.342	0.333	0.303	0.263	(40)	
(41)		taud	2.254	2.313	2.372	2.338	2.408	2.460	2.409	2.426	2.497	2.442	2.318	2.189	(41)	
(42)		Ebn at Noon	576	723	818	839	869	883	849	829	804	697	536	461	(42)	
(43)		Edh at Noon	46	70	87	106	105	102	105	96	77	62	44	36	(43)	
(44)	All-Sky Solar Radiation	RadAvg	0.36	0.98	2.48	3.98	5.14	5.66	5.08	4.07	2.76	1.26	0.46	0.25	(44)	
(45)		RadStd	0.06	0.17	0.28	0.49	0.47	0.28	0.62	0.42	0.38	0.21	0.09	0.03	(45)	

Historical Trends

		DBAvg		Heating		Cooling			Degree-Days			
				99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(46)
(47)	Regional (54 neighbors)	+0.42	N/A	N/A	N/A	N/A	N/A	-128	-156	N/A	N/A	(47)

Nomenclature: See separate page